

[Aim: 100/100 in Maths]

आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #6

#ATK?

PLEQV

→ Word Problems V.V. 2mlg.

#LP: There are two classrooms, (A) and (B). If 5 students are shifted from Room A to Room B, the resulting number of students in the two rooms becomes equal. If 5 students are shifted from Room B to Room A, the resulting number of students in Room A becomes double the number of students left in Room B. Find the number of students in each room.

let room A $\rightarrow$ 'x' students
room B $\rightarrow$ 'y' students

ATQ

$$A' = (x - 5)$$

$$B' = (y + 5)$$

$$A' = B'$$

$$x - 5 = y + 5$$

$$\boxed{x - y = 10} \text{--- (I)}$$

ATQ

$$A'' = (x + 5)$$

$$B'' = (y - 5)$$

$$A'' = 2B''$$

$$(x + 5) = 2(y - 5)$$

$$x + 5 = 2y - 10$$

$$\boxed{x - 2y = -15} \text{--- (II)}$$

① $\Rightarrow$ $x - y = 10$
 $x - 2y = -15$
 $\quad \quad \quad + \quad \quad +$
 $\hline y = 25$

$\Downarrow$

$$x - 25 = 10$$

$$x = 25 + 10$$

$$x = 35$$

Final Ans:

$$\left[\begin{array}{l} A \rightarrow x \Rightarrow 35 \text{ students} \\ B \rightarrow y \Rightarrow 25 \text{ students} \end{array} \right]$$

Fraction Wale Sawaal:

#LP: A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator, and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction. [CBSE 2020]

Let the fraction is

$$\frac{x}{y}$$

numerator
denominator

Ans

$$\Rightarrow \frac{x-1}{y} = \frac{1}{3}$$

$$\Rightarrow 3(x-1) = y$$

$$\Rightarrow 3x - 3 = y$$

$$\Rightarrow 3x - y = 3 \quad \text{--- (I)}$$

$$\Rightarrow \frac{x}{y+8} = \frac{1}{4}$$

$$\Rightarrow 4x = y + 8$$

$$\Rightarrow 4x - y = 8 \quad \text{--- (II)}$$

$$\text{(I)} \rightarrow 3x - y = 3$$

$$\text{(II)} \rightarrow 4x - y = 8$$

$$x = 5$$

$$\text{put } x = 5$$

$$\Rightarrow 4(5) - y = 8$$

$$\Rightarrow 20 - y = 8$$

$$\Rightarrow 20 - 8 = y$$

$$y = 12$$

Final Ans
 $\frac{5}{12}$

#LP: If we add 1 to the numerator and subtract 1 from the denominator, a fraction reduces to 1. It becomes 1/2 if we only add 1 to the denominator. What is the fraction?

let fraction is $\frac{x}{y}$.

ATQ

$$\Rightarrow \frac{x+1}{y-1} = 1$$

$$\Rightarrow x+1 = y-1$$

$$\Rightarrow x - y = -2 \quad (1)$$

$$\Rightarrow \frac{x}{y+1} = \frac{1}{2}$$

$$\Rightarrow 2x = y+1$$

$$\Rightarrow 2x - y = 1$$

$$\frac{x}{y} = \frac{1}{2}$$

final Ans:-

$$\frac{x}{y} = \frac{1}{2}$$

#LP: The numerator of a fraction is 4 less than the denominator. If the numerator is decreased by 2 and the denominator is increased by 1, then the denominator is eight times the numerator. Find the fraction.

Let the fraction is $\frac{x}{y}$

ATQ

$$x = y - 4$$

$$x - y = -4 \quad \text{--- (I)}$$

$$x =$$
$$y =$$

$$\frac{x-2}{y+1}$$

$\swarrow N'$
 $\nwarrow D'$

ATQ

$$D' = 8N'$$

$$y+1 = 8(x-2)$$

$$y+1 = 8x-16$$

$$1+16 = 8x-y$$

$$8x - y = 17 \quad \text{--- (II)}$$

Final Ans

$$\frac{x}{y} =$$

Part Wale Sawaal:

#LP: A part of monthly hostel charges is fixed and the remaining depends on the number of days one has taken food in the mess. When student A takes food for 20 days, she has to pay ₹1000 as hostel charges whereas a student B, who takes food for 26 days, pays ₹1180 as hostel charges. Find the fixed charges and the cost of food per day.

let fix. charge is ₹ x .

per day food charge ₹ y .

(A) → 20 days food
& Total = ₹1000

$$x + 20y = 1000 \quad \text{--- (I)}$$

(B) = ₹1180

$$x + 26y = 1180 \quad \text{--- (II)}$$

$$-x + 20y = 1000$$

$$6y = 180$$

$$y = ₹30$$

$$x + 20(30) = 1000$$

$$x + 600 = 1000$$

$$x = 300$$

Final An

fix → ₹300

Food → ₹30
per day

#LP: The taxi charges in a city consist of a fixed charge together with the charge for the distance covered. For a distance of 10 km, the charge paid is Rs 105 and for a journey of 15 km, the charge paid is Rs 155. What are the fixed charges and the charge per km? How much does a person have to pay for travelling a distance of 25 km?

let fix charge $\Rightarrow \text{₹ } x$
per km charge $\Rightarrow \text{₹ } y$

$$\begin{cases} x = \text{₹ } 5 & \leftarrow \text{fix} \\ y = \text{₹ } 10 & \leftarrow \text{per km.} \end{cases}$$

Ans

$$x + 10y = 105 \quad \text{--- (I)}$$

$$x + 15y = 155 \quad \text{--- (II)}$$

(ii) 25 km??

$$\Rightarrow x + 25y$$

$$\Rightarrow 5 + 25(10)$$

$$\Rightarrow 5 + 250 = \text{₹ } 255$$

#LP: A bookstore shopkeeper gives books on rent for reading. He has a variety of books in his store related to fiction, stories, quizzes etc. He takes a fixed charge for the first two days and an additional charge for subsequent days. Amruta paid ₹22 for a book and kept it for 6 days; while Radhika paid ₹16 for keeping the book for 4 days. Assume that the fixed charge is ₹x and the additional charge (per day) is ₹y. Based on the above information, answer any four of the following questions.

(i) The situation of the amount paid by Radhika. is algebraically represented by (CBSE 2021)

(A) $x - 4y = 16$

(B) $x + 4y = 16$

(C) $x - 2y = 16$

(D) $x + 2y = 16$

Ans (A) $\Rightarrow$ 6 days

$$\boxed{x + 4y = 22} \quad \text{--- (I)}$$

2 days 4 days

let fix charge for first 2 days $\Rightarrow$ ₹x
thereafter per day $\Rightarrow$ ₹y

Ans

(B) 4 days

$$\boxed{x + 2y = 16} \quad \text{--- (II)}$$

2 days 2 days

(ii) The situation of the amount paid by Amruta. is algebraically represented by

- (A) $x - 2y = 11$
- (B) $x - 2y = 22$
- (C) $x + 4y = 22$
- (D) $x - 4y = 11$

(iii) What are the fixed charges for a book?

- (A) ₹ 9
- (B) ₹ 10
- (C) ₹ 13
- (D) ₹ 15

(iv) What are the additional charges for each subsequent day for a book?

- (A) ₹ 6
- (B) ₹ 5
- (C) ₹ 4
- (D) ₹ 3

(v) What is the total amount paid by both, if both of them have kept the book for 2 more days?

- (A) ₹ 35
- (B) ₹ 52
- (C) ₹ 50
- (D) ₹ 58

$$\begin{aligned}x + 4y &= 22 \\x + 2y &= 16 \\ \hline 2y &= 6 \\ y &= 3\end{aligned}$$

put $y = 3$ in $x + 2y = 16$

$$x + 2(3) = 16$$
$$x + 6 = 16$$
$$x = 10$$

$$\begin{aligned}A &\Rightarrow 22 + 2(5) = \underline{28} \\ R &\Rightarrow 16 + 2(3) = \underline{22}\end{aligned} \rightarrow \text{Total} \Rightarrow \boxed{50}$$

INTERST WALE SAWAAL:-

#LP: A man lent a part of his money at 10% p.a. and the rest at 15% p.a. His income at the end of the year is ₹1,900. If he had interchanged the rate of interest on the two sums, he would have earned ₹200 more. Find the amount lent in both cases. (CBSE 2025)

let two parts of money are ₹(x) & ₹(y).

ATQ (case I) $\begin{matrix} x \rightarrow 10\% \\ y \rightarrow 15\% \end{matrix}$

$$x \times \frac{10}{100} + y \times \frac{15}{100} = 1900$$

$$\Rightarrow \frac{10x + 15y}{100} = 1900$$

$$\Rightarrow \boxed{10x + 15y = 190000} \text{--- (I)}$$

$$\begin{matrix} x = \\ y = \end{matrix}$$

(case II)

$$\begin{matrix} x \rightarrow 15\% \\ y \rightarrow 10\% \end{matrix}$$

$$\Rightarrow x \times \frac{15}{100} + y \times \frac{10}{100} = 1900 + 200$$

$$\Rightarrow \frac{15x + 10y}{100} = 2100$$

$$\Rightarrow \boxed{15x + 10y = 210000} \text{--- (II)}$$

#LP: Vijay invested certain amounts of money in two schemes A and B, which offer interest at the rate of 8% per annum and 9% per annum, respectively. He received ₹1,860 as the total annual interest. However, had he interchanged the amounts of investments in the two schemes, he would have received ₹ 20 more as annual interest. How much money did he invest in each scheme? (CBSE 2025)

Let amount invested in (A) $\rightarrow$ ₹ x

(B) $\rightarrow$ ₹ y

ATQ

A $\rightarrow$ 8%

B $\rightarrow$ 9%

$$x \times \frac{8}{100} + y \times \frac{9}{100} = 1860$$

$$\Rightarrow \frac{8x + 9y}{100} = 1860$$

$$\Rightarrow 8x + 9y = 186000 \quad \text{--- (2)}$$

ATQ (A) $\rightarrow$ ₹ y (B) $\rightarrow$ ₹ x

A $\rightarrow$ 8%

B $\rightarrow$ 9%

$$\Rightarrow y \times \frac{8}{100} + x \times \frac{9}{100} = 1860 + 20$$

$$\Rightarrow \frac{8y + 9x}{100} = 1880$$

$$\Rightarrow 8y + 9x = 188000 \quad \text{--- (1)}$$



HW

→ Clear notes**THANK YOU COODIESS !!**